

THE auto-CAD function — what it does, how it works, how to make it better

Turns every catalogued part into a shaded, dimensioned CAD image from its name + NSN + FLIS dimensions. Offline, pure Python + Pillow.

WHAT IT DOES

Most of the 20,869 representative parts have no manufacturer CAD model and often no manual figure — just a name and FLIS dimensions. The auto-CAD function manufactures a recognizable engineering picture for each part's shape, scales it to the real dimensions, and renders a clean three-quarter CAD view with dimension callouts and a title block. The result looks like a drafting-room approximation and is labeled as such — a fast way to get a part that otherwise had none.

HOW IT WORKS · the pipeline (engine/cad_render.py)

1 CLASSIFY

- name -> family() (24 keyword rules)
- if unknown -> NSN Federal Supply Class
- 22 shape families (bolt..canister)

2 DIMENSION

- parse FLIS characteristics text
- DIAMETER/LENGTH/WIDTH/HEIGHT/THK
- > L, W, H, dia, bore (inches)

3 BUILD MESH

- family builder -> {V, F}
- primitives: cyl/prism/tube/torus/
- sphere/helix/gear/box, scaled

4 PROJECT

- rotate (three-quarter iso)
- orthographic to 2-D
- fit + centre to canvas

5 SHADE

- painter's sort by depth
- per-face normal . light
- facet colour = material x brightness

6 ANNOTATE

- overall-dimension callouts (in.)
- title block: NSN / name / shape
- drafting grid + 'REPRESENTATIVE'

7 OUTPUT

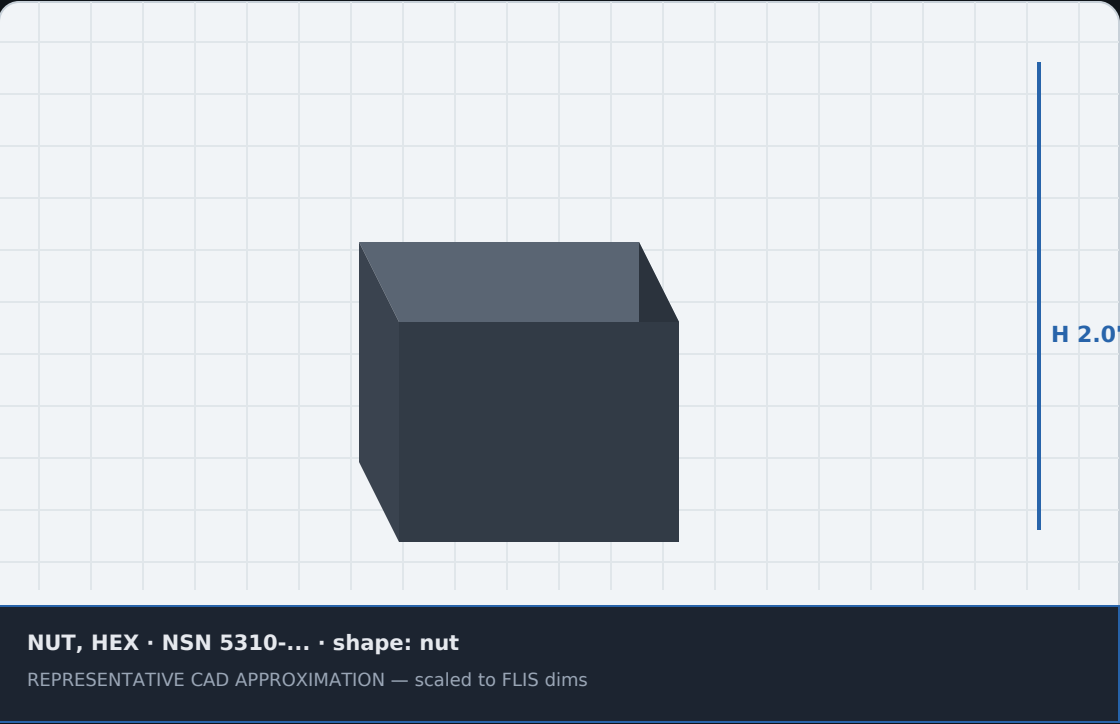
- 2x supersample -> LANCZOS down
- PNG to index/cadcache/ sidecar
- cache key = nsn + CAD_VERSION

8 SERVE

- /cadimg?nsn= render-on-demand
- MAKE-CAD.bat pre-renders all
- shown as the 3-D card thumbnail

Faithful to partgeo.js: the same 22 families + the same NSN/FSC fallback, so the CAD image always matches the live WebGL model and the card tag.

ANATOMY OF A CAD IMAGE



- Shaded facets**
per-face flat shading from a key light
- Edge lines**
every face outlined — the 'CAD wireframe' read
- Dimension callout**
the overall size in inches, from FLIS
- Drafting grid**
faint background = engineering-paper feel
- Title block**
NSN · name · shape family · disclaimer

BETTER & MORE — ways to grow this feature

Higher fidelity

- smooth (vertex-normal) shading for round parts
- port the WebGL specular/metallic look to Pillow
- soft shadow + ground plane for depth
- per-family canonical orientation (bolt head up, ...)

More accuracy

- parse thread pitch, head/drive type, # holes
- sub-type builders (hex vs socket screw; ball vs roller)
- fit dims from RPSTL/figure when FLIS is silent
- 'low-confidence' badge when dimensions are defaulted

Real CAD output

- export STL / OBJ per part (3-D print / CAD)
- glTF/GLB for web + AR viewing
- true multi-view drawing sheet (front/top/side + iso)
- full dimensioning + tolerances on the sheet

UX + scale

- CAD tab + download button in the part modal
- loupe on the CAD image; print into the job packet
- 'report wrong shape' override -> append-only feedback
- multiprocess the 20,869 batch; pre-warm on idle